

Playwright Automation Framework Guide

Production-oriented keyword-driven framework documentation

Playwright

Keyword Driven

Excel Based

PDF Validation

SUPPORTED FIELD TYPES

12

ACTION KEYWORDS

10+

CORE LAYERS

3

DATA SOURCES

Excel + .env

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1) Introduction

This framework is a **keyword-driven Playwright automation framework** where test steps are authored in Excel sheets and executed through reusable action handlers.

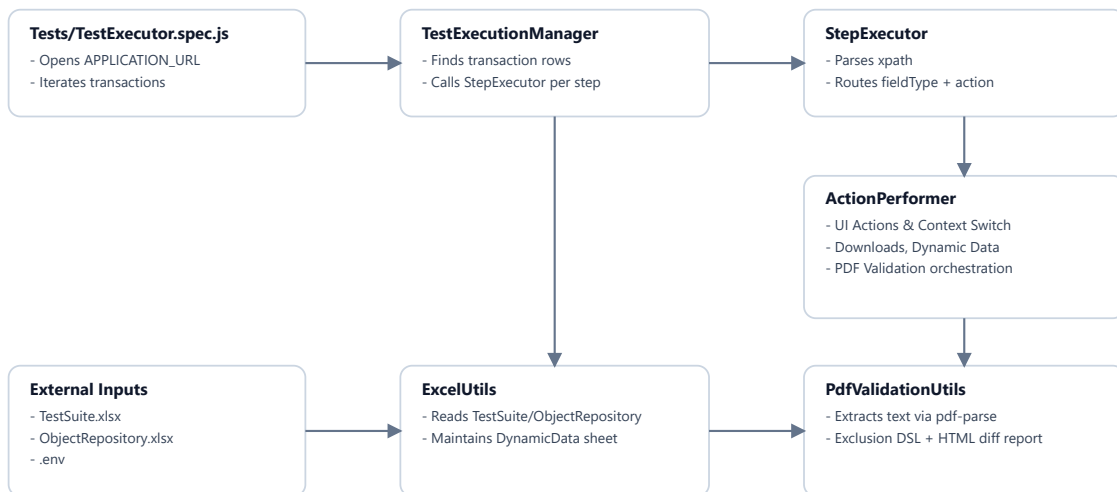
It is designed for both automation engineers and business users by separating **test intent (what to do)** from **technical implementation (how to do)**.

2) Framework Capabilities

- Data-driven transaction execution from TestSuite and ObjectRepository.

- Supports common UI actions (input, click, validation, dropdown, window/frame switch).
- Dynamic data read/write support for cross-step data management.
- Download handling and advanced PDF comparison with HTML diff reports.
- Config-driven behavior using `.env` (URLs, paths, report settings, exclusions).

3) Architecture Diagram



Legend: The framework takes data from Excel + `.env`, routes steps through execution layers, and generates actionable reports.

4) Execution Flow (Standard Procedure)

1. Read test cases from TestSuite control sheet and filter executable rows.
2. Open browser using `APPLICATION_URL`.
3. Resolve each transaction to corresponding TestSuite and ObjectRepository sheets.
4. For each step, map `fieldType` + `action` to the exact performer method.
5. Execute UI/API/data actions; generate Playwright + custom reports for evidence.

5) Field Types and Associated Actions

Text Box

ACTION	DESCRIPTION	TEST DATA USAGE
I	Inputs text into a textbox locator.	Enter the text value.

Button

ACTION	DESCRIPTION	TEST DATA USAGE
C	Clicks a button or clickable element.	Any non-null value to trigger action.

Checkbox

ACTION	DESCRIPTION	TEST DATA USAGE
C	Checks/Unchecks based on test data.	Use exactly Check or Uncheck.

Label

ACTION	DESCRIPTION	TEST DATA USAGE
V	Validates locator text equals expected.	Expected visible text.

Dropdown

ACTION	DESCRIPTION	TEST DATA USAGE
I	Selects by option value.	Option value/string.
INDEX_SELECTION	Selects dropdown item by index.	Numeric index (single digit in current logic).

SwitchToIframe

ACTION	DESCRIPTION	TEST DATA USAGE
SWITCH	Switches context into iframe.	Any non-null trigger value.

SwitchToWindow

ACTION	DESCRIPTION	TEST DATA USAGE
SWITCH_WINDOW	Clicks locator and switches to newly opened window/tab.	Any non-null trigger value.

SwitchToMainWindow

ACTION	DESCRIPTION	TEST DATA USAGE
SWITCH	Switches back to parent/main page context.	Not required.
SWITCH_BY_TITLE	Switches to window by title.	Provide target window title.

DynamicDataHandler

ACTION	DESCRIPTION	TEST DATA USAGE
W; ;<COLUMN_NAME>	Writes extracted UI text to dynamic data sheet.	Uses current locator text content.

PdfValidation

ACTION	DESCRIPTION	TEST DATA USAGE
V	Downloads PDF, compares with expected PDF, applies exclusion DSL, generates HTML diff report and attaches artifacts.	Expected file name (with/without .pdf).

API

ACTION	DESCRIPTION	STATUS
GET	Reserved for API GET execution path.	Placeholder

OCR

ACTION	DESCRIPTION	STATUS
N/A	OCR field type hook exists for future implementation.	Placeholder

6) TestSuite Structure

Control Sheet: Used to list scenarios, execution flag, and transactions.

COLUMN	MEANING
A	TestCaseName
B	Category
C	Execute (Y/N)

COLUMN	MEANING
D onward	Transaction names to run in sequence

Transaction Sheets: Row 1 = TestCase IDs across columns, Row 2 = Transaction names, Row 3 onward = step data with Field Name in column A.

7) ObjectRepository Structure

COLUMN	MEANING
A	Field Name (must match TestSuite field name)
B	Field Type (e.g., Text Box, PdfValidation)
C	Action keyword (e.g., I, C, V, SWITCH)
D	Xpath / locator expression

8) Sample Data Structures

Sample TestSuite (Transaction Sheet)

FIELD NAME	TC_PDF_001 / TXN_PDF_VALIDATE
Download Brochure	Get_Started_With_Smallpdf.pdf
Customer Name	John Doe
Invoice Date	12/04/2026

Sample ObjectRepository (Same Sheet Name)

FIELD NAME	FIELD TYPE	ACTION	XPATH
Download Brochure	PdfValidation	V	//a[@id='download-pdf']
Customer Name	Text Box	I	//input[@name='customerName']

FIELD NAME	FIELD TYPE	ACTION	XPATH
Invoice Date	Label	V	//span[@id='invoiceDate']

9) Environment Configuration (.env)

VARIABLE	PURPOSE	EXAMPLE
APPLICATION_URL	Application base URL for test navigation	https://example.com
DOWNLOAD_FOLDER_PATH	Target folder for downloaded PDFs and reports	F:/PlaywrightTests/AutomationFramework/project/test/downl
EXCLUDE_CONTENT	Custom DSL for dynamic text exclusions in PDF comparison	WORD:John;;DATE:ANY;;REGEX:/Invoice\\s*No\\s*:\\s*\\w+/i

Tip Keep expected files under DOWNLOAD_FOLDER_PATH/expected.

10) Documentation & Usage Standards

- Use consistent naming between TestSuite Field Name and ObjectRepository Field Name.
- Use explicit action keywords and valid field types only.
- Store expected PDFs in `expected` folder with stable naming convention.
- For dynamic documents, use `EXCLUDE_CONTENT` DSL to avoid false mismatches.
- Attach generated reports as part of execution evidence in Playwright HTML report.